Judging the burn-scar labels

The project file is `data/labels/biak_labels.qgz`. Open it in QGIS 3.44 LTR, not 4.2.
These labels become the reference that the Task 18 burned-area product is measured against (Task 22). Nothing else in this project can do that: the field survey was stopped by security conditions, and F18 showed that no prompted segmentation model can stand in.

What you are looking at

layer	what it is
LABELS — draw here	142 boxes of 60 x 60 m, already placed; you fill in the class
samples	the 42 random points each box is centred on
VIIRS footprints	375 m circles round the detections — where the samples were drawn from, not truth
VIIRS/MODIS detections	the individual satellite detections
S2DR4 1 m imagery	SWIR and true colour, three dates per event
Sentinel-2 20 m	before the event, 23 August, 28 August

Only the 1 m SWIR image of 28 August for E0313 is switched on at the start. Tick the others as you go.
There is no dNBR+ layer, on purpose. These labels judge that product, so seeing it while judging would make the comparison worthless. Do not add it.

The dates, and why there are three

All three events ran until 25 August, so no single image shows the whole story.

date	1 m	20 m	what it shows
before the event	no	yes	the ground as it was, two to twelve weeks earlier
23 August	yes	yes	mid-episode, the clearest scene of the run, and the field survey's basemap
25 August	yes	no	the last detection day
28 August	yes	yes	after, and the image dNBR+ compared against

A fresh scar reads bright orange on 23 August and fades to dull brown by 28 August. That fading is the most reliable sign you have. Ground that was already open before the event looks the same on all three dates, and that is the difference between burnt and merely bare.
So the question to answer for each box is not "does this look burnt" but **"did this change between the dates, and does the change look like burning"**. dNBR+ measures change between two looks; judging one date alone asks something different, and the two then disagree for reasons that have nothing to do with accuracy.

The imagery

SWIR (B12/B8A/B4) is the one to judge from. In it:

what you see	what it is
bright green	healthy vegetation, the greener the denser
brown to maroon, dull red	burnt or bare ground — this is what you are looking for
pale green to cream	grass, dry or sparse cover
near black	water
white, sharp edges	roofs and paved surfaces

Switch to true colour when you want to check what a feature is — a roof, a track, a bare field — because true colour looks like a photograph and SWIR does not.
The three channels share one fixed brightness scale, so these colours mean the same thing in every tile and can be compared between events. (An earlier version of this imagery

stretched each channel separately, which showed vegetation orange and burnt ground blue; if you see that, the imagery is stale — re-run `scripts/label_prepare.py`.)
The 1 m detail is super-resolved, which means a model generated it from 10 m pixels. Trust the shape of a patch and the contrast between patches. Do not trust a single sharp edge or a small isolated object: those are the parts a super-resolution model invents.

The job: judge the boxes

There is nothing to draw. The label layer holds **142 boxes of 60 x 60 m**, 0.36 ha each, which is 9 pixels of the 20 m product being tested. Your job is the judgement only: what is inside each box. The 75 new boxes (`MAP26` – `MAP50` , `B01` – `B50`) are the ones still to do.
Why fixed boxes rather than outlines: the sample location has to come from the random draw. Choosing where to draw, by eye, around a scar that is visible turns the reference into a collection of obvious cases, and the agreement figure then measures the easy ones only.

Two strata, and why

`sampling`	how many	drawn from
`random_box`	42, judged	random points, 7 inside and 7 outside each event's VIIRS footprint
`map_burned`	50: 25 judged, 25 to do	random points inside the dNBR+ polygons of the same three events
`footprint_unmapped`	50, to do	random points inside the VIIRS footprints, on ground the polygons do not claim

The first 42 produced only 3 burned boxes, which is too few to say anything: the interval on 3 of 8 runs from about 14% to 69%. The second stratum samples the map being tested, so the burned class is well represented and the two strata together give a corrected area estimate with a confidence interval.
The two drawn strata were raised to 50 boxes each in Task 23: at 20 and 10 judged boxes the 95% interval on the burned area was too wide to say whether the map's own figure sat inside it, and which estimator you used decided the answer. Fifty per stratum is the floor Olofsson et al. 2014 recommends for a change stratum. The 42 original `random_box` boxes keep their place and are still used, but only for the stratum their own draw supports; the `footprint_unmapped` boxes are a separate design and are not mixed with them.
Sampling by the map is not circular, because the map is never shown while judging. Judging by the map would be. That is why there is still no dNBR+ layer in the project.

1. Open the project, select **LABELS — draw here**, open the attribute table (**F6**).
2. Toggle editing (**Ctrl+E**).
3. Click a row, then the **Zoom to feature** button, so the map shows that box.
4. **Look at the dates in turn**, which is the part that decides the answer:
5. 1 m SWIR, 23 August: is the box orange, meaning freshly burnt?
6. 1 m SWIR, 28 August: has it gone dull brown, as a scar does in five days?
7. 20 m before: was it already open and bare before the event?
8. Fill two columns:
9. `class` — `burned` , `unburned` or `unsure` ;
10. `confidence` — `high` , `medium` or `low` ;
11. `notes` — only if something needs saying.
12. Save (**Ctrl+S**) often.

The table also has a **form view** at the bottom right, which shows one box at a time if that reads more comfortably than the grid.
Spelling matters, and `scripts/label_check.py` catches anything unexpected.

What each pattern means

before	23 Aug	28 Aug	read it as
green	orange	brown	burnt during the episode, the clearest case
green	green	brown	burnt between 23 and 25 August
brown or bare	same	same	open ground already, not a scar
green	dark patch	green	cloud or its shadow, not a scar

If a box straddles two cover types

Judge what covers most of it. If it is close to half and half, that is exactly what `unsure` is for.

The rules that keep this honest

- **`unsure` is a real answer.** Thin smoke, cloud shadow, a patch you would argue about —
- mark it `unsure` and move on. Task 22 drops those and reports how many there were. A
- guessed label is worse than no label, because it silently moves the accuracy figure.
- **Do not move a box** to somewhere easier to judge, and do not resize it. The points were
- drawn at random from a fixed seed; shifting them onto convenient ground is exactly the
- bias the random draw exists to prevent. A box on water, on a roof or under cloud gets
- `unsure` and a note.
- **Inside the footprint does not mean burned.** The footprint only says a satellite saw
- heat within 375 m at some point. Plenty of ground inside it did not burn, and that is
- precisely what the labels are meant to establish. Judge each patch on the imagery.
- **Outside the footprint does not mean unburned** either. If it looks burned, label it
- burned.
- **Do not use the detections as evidence.** They tell you where heat was seen, which is
- the very thing the product is trying to convert into area. Use them to orient yourself,
- not to decide.
- **Change is the question, not appearance.** Bare ground that looks the same on every date
- is `unburned`, however brown it is. That distinction is the whole reason three dates are
- in the project.
- **Re-judging a box is allowed, and expected.** The first pass was made on one date only.
- If a box reads differently with the sequence in front of you, change it and say why in
- `notes`. Nothing is gained by defending an earlier call.

When you are done

```
python scripts/label_check.py
```

It reports how many polygons exist, whether each holds its own sample point, the class counts by stratum, the polygon sizes, and anything malformed or misspelled. It changes nothing.

Partial work is fine — the check tells you what is left. Task 22 can run on fewer than 42 polygons, it will just say so and carry a wider uncertainty.

The three scripts, in the order you need them

script	when	what it does
`scripts/label_prepare.py`	already run	builds the imagery for every downloaded date, the sample points and the 42 boxes
`scripts/label_pre_imagery.py`	already run	pulls the 20 m before, mid and after chips from Earth Engine
`scripts/label_add_boxes.py`	when a stratum needs more boxes	draws more `map_burned` or `footprint_unmapped` boxes at random, at least 150 m from any existing one
`scripts/label_fill_ids.py`	only if you add polygons by hand	fills `sample_id` and `event_id` from the point inside each polygon
`scripts/label_second_pass.py`	once the first pass is done	copies the boxes into a fresh file for a second interpreter, then compares the two
`scripts/label_check.py`	any time	reports what is there and what is wrong

`scripts/label_qgis_project.py` rebuilds the `.qgz` itself. It never touches the GeoPackage, so polygons already drawn survive a rebuild.

The second interpreter

Every agreement figure in the project so far comes from one person reading one set of images. That is the largest unmeasured uncertainty there is, larger than the choice of interval, so the boxes are judged a second time, blind.

```
python scripts/label_second_pass.py           # build the second filepython scripts/label_second_pass.py --compare  # kappa, agreement, disag
```

The first command writes `data/labels/biak_labels_pass2.gpkg`: the same boxes, the same imagery, with `class`, `confidence` and `notes` emptied. Nothing of the first pass is in that file, so the second interpreter cannot see the first answers. Open the **pass-2** GeoPackage in a new QGIS project (`label_qgis_project.py` builds only the first one), judge the boxes the same way, and save.

Run the compare once both are filled in. It reports Cohen's kappa and the raw agreement, per stratum, and lists every box the two readers disagreed on with both answers. **Neither answer wins automatically.** The disagreement rate is the result; deciding a reconciled label is the owner's job, box by box, with a reason recorded.